PRESCRIPTIVE ANALYTICS-BASED SYSTEM FOR BORROWERS RISK AND POVERTY ASSESSMENT WITH SCHOLARSHIP ALLOCATION FOR CARD, INC.

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MASTER IN INFORMATION TECHNOLOGY

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APPROVAL SHEET

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DEDICATION

This study is dedicated to my family, whose love, support, and sacrifices have always been my source of strength and inspiration. Their constant encouragement and belief in me helped me push through the challenges and finish this journey.

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ABSTRACT

This study addresses the lack of an integrated, data-driven system for borrower risk assessment, poverty evaluation, and scholarship allocation in the selected areas in CARD, Inc. It aims to develop a prescriptive analytics-based system that combines predictive modeling and rule-based decision support to improve operational efficiency and decision-making. Specifically, the study seeks to predict borrower risk and Progress Out of Poverty Index (PPI) likelihood and design a scholarship prioritization framework based on client vulnerability and dependent information. The study employed an experimental research design using real client datasets, including loan records, repayment behavior, PPI scores, and dependent profiles. Machine learning algorithms such as Linear Regression, Logistic Regression, Random Forest, and Gradient Boosting were used for predictive modeling, while a rule-based approach was applied for scholarship prioritization. Results show that Gradient Boosting and Linear Regression achieved the highest predictive performance, effectively identifying at-risk borrowers and PPI likelihood. The system demonstrated efficient performance, high usability, and positive user acceptance, with all test cases passing successfully. Findings indicate that integrating predictive and prescriptive analytics improves accuracy, consistency, and decision-making processes. The study concludes that the developed system is functional, reliable, and suitable for supporting CARD, Inc.'s operations. It enhances risk identification, poverty monitoring, and equitable scholarship allocation, contributing to more effective poverty alleviation efforts.

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DEFINITION OF TERMS

The following terms were conceptually and operationally defined for the purpose of the research study.

Algorithm	A method for resolving a well-defined computational problem.
Data Set	A set of raw data made up of individual sounds that will be used to train the model.
Model	The process of creating a descriptive diagram of the relationships between different types of data that will be stored in a database

ACRONYMS

CARD, Inc.	Center for Agriculture and Rural Development (CARD), Inc. (A Microfinance NGO)
PPI	Progress out of Poverty Index
RAD	Rapid Application Development
PSSUQ	Post-Study System Usability Questionnaire
TAM	Technology Acceptance Model
PU	Perceived Usefulness
PEOU	Perceived Ease-of-Use
AUC	Area Under Curve
CA	Classification Accuracy
CRISP-DM	Cross-Industry Standard Process for Data Mining

CHAPTER I

INTRODUCTION AND ITS BACKGROUND

The use of analytics in microfinance institutions (MFIs) continues to grow, enhancing borrower risk prediction and portfolio quality assessment. Research shows that predictive and prescriptive analytics improve credit scoring, reduce portfolio-at-risk and help detect borrower vulnerabilities. Data analysis has greatly enhanced Microfinance Institutions' methods for mitigating risk (Abusharbeh, 2023). Additionally, advanced predictive models allow for an increased level of detail in assessing and segmenting borrower risks (Bone-Winkel & Reichenbach, 2024). Beyond risk assessment, data analytics offers valuable insights into borrower behavior and vulnerability, which is essential for preventing excessive indebtedness and supporting responsible lending practices (Kasoga & Tegambwage, 2021). Overall, the findings from these studies indicate that data analytics is a major contributor to developing sustainable lending practices in microfinance institutions.

In addition to predictive risk assessment, data-driven methods are used to prioritize scholarship awards and assess or measure poverty within microfinance-focused NGOs. Recent study by Fajardo et al. (2024) demonstrates that machine learning classification models can effectively forecast scholarship allocation, thereby optimizing educational resources. The Grameen Foundation developed the Progress out of Poverty Index (PPI) as a country-specific, survey-based tool that uses simple household indicators to estimate the likelihood that a family is living below the poverty line. In the Philippines, the PPI is widely used by

microfinance institutions to assess client poverty levels, monitor outreach to low-income populations, and support data-driven targeting of financial and social services. Studies by Dr. Vandana Sethi (2024) confirm the reliability of poverty scorecards in predicting deprivation and tracking poverty transitions. These tools help microfinance identify vulnerable populations and form the basis for data-driven program planning, including education support, social protection and poverty monitoring. Fonseca et al. (2024) identify analytics-driven poverty assessments will play a key role in developing sustainable poverty reduction strategies.

The integration of these analytical methods aligns with the United Nations Sustainable Development Goals, particularly SDG 1: No Poverty. Microfinance is recognized as an effective tool for reducing poverty, strengthening household resilience and empowering women through socio-economic advancement (Gondraneos et al., 2025; Almase, 2021; Ashraf et al., 2024). Recent studies show that data-driven microfinance interventions produce more equitable outcomes and enhance poverty reduction in various contexts (Lahnech & Chami, 2025). The use of borrower risk prediction, poverty assessment and the prioritization of scholarships will support SDG1 by creating two mechanisms to reduce poverty and improve economic shock resistance.

Advancements in technology have allowed for faster adoption of analytics frameworks into the microfinance industry. Descriptive, predictive and prescriptive analytics now leverage data streams (Almeida et al., 2023), clustering to segment borrower behavior (Chaudhry et al., 2023; Alam et al., 2021), deep learning for

time series forecasting (Li & Law, 2024) and geospatial analytics to identify socioeconomic disparities (Moos et al., 2021). All of these technologies allow MFIs to track the behaviors of borrowers, assess their poverty risks and identify geographic areas where there are educational needs more effectively. Additionally, Geospatial Big Data (Al-Yadumi et al., 2021) and Business Intelligence Visualization Tools (Shubho et al., 2022) provide MFIs with the tools to make informed decisions on how to allocate resources and target programs based upon evidence.

These analytical capabilities support the development of integrated support system, "Prescriptive Analytics-Based System for Borrowers Risk and Poverty Assessment with Scholarship Allocation" for Center for Agriculture and Rural Development (CARD, Inc.) (A Microfinance NGO). The prescriptive analytics model is applied to develop a single support system using borrower risk prediction, PPI-based poverty assessment and scholarship prioritization for the benefit of CARD, Inc. The advanced data modeling capabilities of this system allow it to identify which borrowers are at the highest risk of defaulting, identify those clients who are either in or approaching poverty and determine scholarship needs based on the clients' PPI score and risk classification. Using these types of capabilities allows CARD, Inc. to advance its mission to alleviate poverty, to maintain a sustainable loan portfolio, to serve the poorest populations and to better allocate geographically based scholarships. By combining descriptive, predictive and prescriptive analytics, CARD, Inc. can use a data-driven decision-making process

to promote the welfare of clients, to produce better results from programs and to continue its commitment to reducing poverty in the Philippines.

Research Problem

Although CARD, Inc. is committed to reducing poverty, it lacks an evidence-based system to identify high-risk borrowers or accurately assess whether clients remain in poverty using the Progress out of Poverty Index (PPI). This lack of analytical tools limit CARD, Inc.'s ability to identify at-risk borrowers early, monitor changes in poverty status and allocate scholarship funds to those most in need. As a result, operational blind spots reduce the efficiency of CARD, Inc.'s microfinance and social development programs.

Specific problems to be addressed by the proposed system include:

- How can the system predict borrower risk using the available client and loan data of CARD, Inc.?
- How can the Progress out of Poverty Index (PPI) be integrated into an automated system for more efficient assessment of borrower's poverty status?
- How can the proposed system improve operational efficiency in managing client risk and scholarship prioritization in CARD, Inc.?
- How can the system integrate different datasets (loan records, PPI scores, dependents information) into a single platform for easier analysis and reporting?

The need of addressing these issues is evident in microfinance and poverty research. Studies show that microfinance lacking risk prediction methods face higher portfolios-at-risk and less efficient credit operations (Abusharbeh, 2023; Kasoga & Tegambwage, 2021). Research also indicates that poverty measurement systems such as PPI help identify vulnerable households and track changes in income levels (Sethi, 2024). Additionally, when scholarship programs do not have sufficient structural analysis, resources are distributed inequitably and produce fewer positive outcomes (Fajardo et al., 2024). Although CARD, Inc. uses experience-based evaluations, the absence of objective analytical methods can result in inconsistent risk evaluations, delayed poverty monitoring and scholarships may not reach the most vulnerable students. These findings highlight the urgent need for a formal, data-driven approach.

To address these needs, this study recommends developing a Prescriptive Analytics-Based System for Borrowers' Risk and Poverty Assessment and Scholarship Allocation for CARD, Inc. This system would integrate predictive modeling, PPI-based poverty scoring and data-driven scholarship prioritization into a single decision support platform.

Research Objectives

General Objective

The study aims to develop an integrated system for CARD, Inc. that combines poverty assessment, scholarship prioritization and predictive modeling

to accurately identify borrower risk and support informed decision-making within the organization.

Specific Objectives:

Specifically, the study aims to:

1. Develop a predictive model to identify borrowers at high risk of default based on their loan information and payment behavior.
2. Develop a system capable of mapping CARD, Inc. clients using their Progress Out of Poverty Index scores.
3. To develop a vulnerability monitoring dashboard that tracks changes in clients' poverty profiles over time.
4. Design a scholarship prioritization module based on the client's PPI score and risk classification.
5. Evaluate the developed system's predictive accuracy, operational performance, and user usability through model evaluation metrics, system testing procedures, and usability assessment methods.

Research Framework

This section presents the theoretical framework that guide the design, development and the implementation of the Prescriptive Analytics-Based System for Borrowers' Risk and Poverty Assessment with Scholarship Allocation in CARD, Inc. The theories are arranged according to their direct application to the system, beginning with those theories that support analytical modeling, followed by decision-making theories, human-development perspectives and adoption of new

technology by CARD, Inc. Collectively, these frameworks provide a foundation for the technical feasibility of the system, its social relevance and operational feasibility for CARD, Inc.

Theoretical Framework

Artificial Intelligence (AI) and Machine Learning (ML) provide the technical foundation for the system's predictive analytics. These allow computational models to find patterns, discover relationships and classify data results from large, complex datasets. According to Gibson et al. (2023), AI-based predictive analytics frameworks can increase the accuracy of analytical results through adapting to changes in the data set and improving predictive accuracy over time. This adaptability is crucial for microfinance institutions, where borrower behavior and socio-economic conditions often shift.

In credit risk prediction, studies show that machine learning models consistently perform better than traditional rules-based systems by identifying non-linear relationships, grouping similar risk levels together and identifying when a borrower is more vulnerable (Wang et al., 2020; Mhlanga, 2021). Machine Learning is also used to predict poverty and socio-economic deprivation using household indicators (Usmanova et al., 2022). These theories provide the framework for developing the predictive capabilities of the system such as borrower risk classification, PPI indicator-based poverty scoring and scholarship prioritization.

The study adopts the **CRISP-DM Framework** as the primary analytics methodology to guide the development of borrower risk prediction and poverty assessment models. The framework consists of six iterative phases, business understanding, data understanding, data preparation, modeling, evaluation, and deployment which collectively ensure the quality, consistency, and reliability of analytical outputs (Saltz, 2021). Studies have shown that CRISP-DM is widely applied in financial and risk analytics because it provides a structured process for comparing, selecting, and validating predictive models and evaluation metrics. Rawat (2023) demonstrated the use of CRISP-DM in credit risk prediction by systematically preparing borrower data, testing multiple machine learning algorithms, and evaluating model performance using predictive accuracy measures. Similarly, Per et al. (2022) emphasized that CRISP-DM enables organizations to compare different machine learning techniques and identify the most effective model for financial decision-making through iterative evaluation and refinement.

Moreover, CRISP-DM supports model prioritization and decision optimization by integrating business objectives with analytical validation. Plotnikova et al. (2022) highlighted that the framework is particularly effective in financial services because it aligns predictive modeling with operational requirements, ensuring that selected models remain interpretable, auditable, and applicable in real-world decision processes. Casonatto et al. (2024) further explained that CRISP-DM strengthens quality and risk management by establishing a systematic approach for assessing model reliability, comparing

evaluation metrics, and minimizing analytical bias. Guided by these principles, the present study utilizes CRISP-DM to evaluate and prioritize models for identifying at-risk borrowers and predicting poverty conditions using indicators such as repayment behavior and Progress out of Poverty Index (PPI) scores. The framework also supports the integration of the selected models into a prescriptive decision-support tool that promotes transparency, replicability, and evidence-based scholarship allocation decisions.

Data-Driven Decision Making (DDDM) establishes that decisions should rely on empirical evidence, structured data and analytic models rather than intuition or subjective judgment (Gill et al., n.d.). In microfinance, depending only on experience-based assessments leads to inconsistencies in evaluation of borrowers, poverty classification and scholarship prioritization.

Incorporating DDDM theory strengthens the study by ensuring that AI/ML model predictions are systematically translated into actionable decisions rather than treated as isolated outputs. Sebestyén (2021) notes that organizational culture, data literacy and user confidence affect DDDM adoption. Therefore, the system must clearly present insights, justify predictions and support staff in making informed decisions. This approach ensures CARD, Inc. uses data effectively and consistently across all branches.

Decision Theory supports the system by providing a framework for making decisions that maximize benefits when the potential outcomes are uncertain. It provides tools for evaluating the probability of outcomes, evaluating alternatives options and selecting the action that maximizes expected benefit or minimizes risk

(White, 2007). Microfinance institutions operate in conditions of uncertainty regarding borrower repayment capacity, fluctuating economic conditions and household vulnerabilities**.**

By grounding the prescriptive system in Decision Theory, the recommendations become more objective, consistent and aligned with CARD, Inc.'s mission to support vulnerable households.

Optimization Theory is central to the system's prescriptive analytics component. While predictive analytics estimates future outcomes, prescriptive analytics recommends the best actions to achieve desired goals under constraints. Optimization models aim to maximize or minimize objective functions—such as minimizing loan default risks or maximizing the equitable distribution of scholarships—while respecting resource limitations (Giesecke et al., 2018).

Wissuchek and Zschech (2025) define prescriptive analytics as the process of turning predictions into actionable recommendations using optimization methods. In this study, Optimization Theory systematically prioritizes scholarship allocation based on poverty severity and borrower vulnerability. It also directs targeted interventions for borrowers at risk of default, enabling CARD, Inc. to identify areas vulnerabilities.

The **Capability Approach**, based on Amartya Sen's work, is a theoretical framework for understanding poverty and educational disadvantage from a human development perspective. The theory views poverty not limited only as low income but as a lack of meaningful opportunities and capabilities needed for individuals to

live fulfilling lives. This perspective aligns with CARD, Inc's mission as a microfinance NGO committed to both financial access and improved well-being.

In this study, the Capability Approach supports the use of the Progress Out of Poverty Index (PPI), as its indicators reflect life conditions, household vulnerabilities, asset deprivation and barriers to opportunity. By grounding poverty scoring and scholarship ranking in this theory, it ensures that the most capability-deprived households are given first consideration.

The final component of the theoretical framework is the **Technology Acceptance Model (TAM)**, which explains how personnel of CARD, Inc. may adopt and utilize the proposed prescriptive analytics system. According to TAM, user acceptance of a system is primarily influenced by two key factors: perceived usefulness and perceived ease of use. In this study, only these two TAM constructs will be utilized in evaluating the system because they are considered the core determinants of user acceptance and intention to use a technology. Previous studies emphasized that perceived usefulness and perceived ease of use are fundamental variables in determining whether users are willing to adopt and continuously use an information system (Lewis, 2019). These constructs are also widely applied in information systems and usability studies because they directly measure how users evaluate the effectiveness and usability of a system in supporting their work-related tasks.

Perceived usefulness refers to the extent to which users believe that the system can improve their work performance, productivity, and decision-making processes, while perceived ease of use refers to the degree to which users

perceive the system as easy to learn, understand, and operate (Lewis, 2019). Lewis (2019) further explained that these constructs remain essential measures in evaluating technology acceptance because they influence users' perceptions and overall interaction with a system. These two constructs are particularly relevant to the study since the proposed prescriptive analytics system is intended to assist personnel in monitoring, analysis, and decision-making activities. Evaluating perceived usefulness will help determine whether the system provides practical value to users, whereas evaluating perceived ease of use will assess whether users can efficiently interact with the system without difficulty or extensive technical knowledge.

By focusing on perceived usefulness and perceived ease of use, the study aims to support the successful adoption of the system and encourage its integration into organizational practices. The use of these constructs will also provide insights into the acceptability of the proposed system among intended users and identify whether the system can meet operational needs effectively and conveniently.

Conceptual Framework

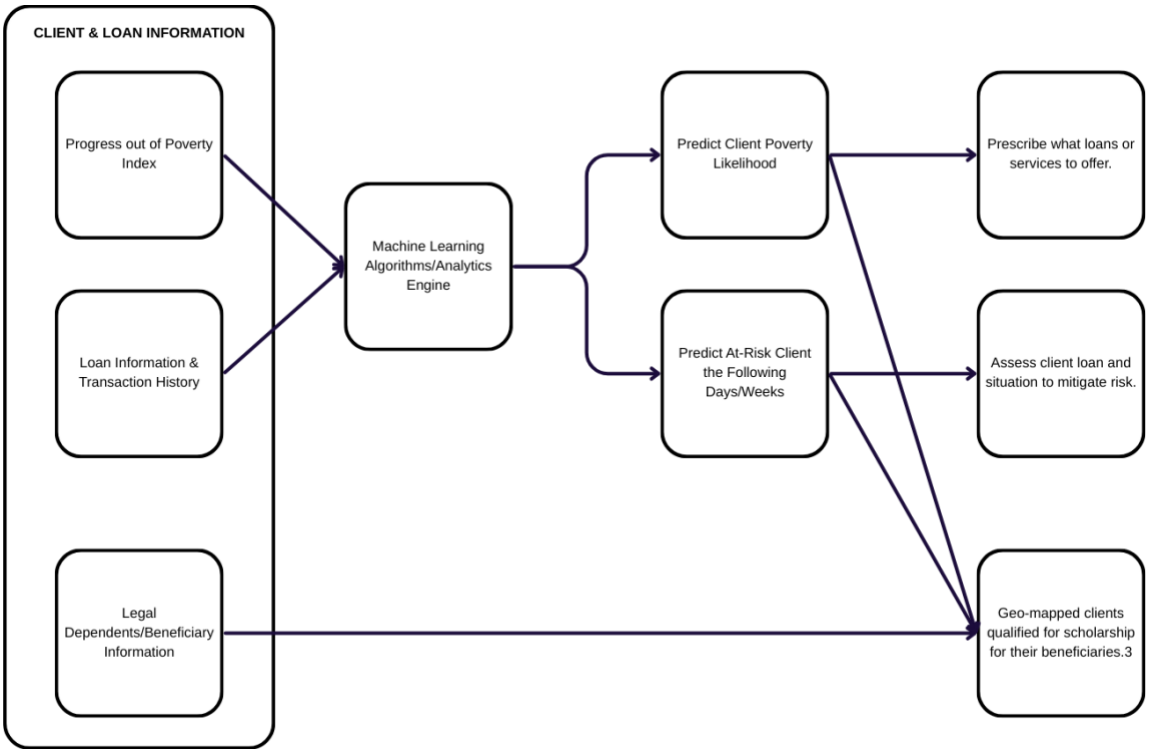
The conceptual framework shows how client and loan data are converted into predictive and prescriptive insights to support financial and poverty-alleviation decisions at CARD, Inc. Three main input variables—Progress out of Poverty Index (PPI) scores, loan and transaction histories and information on legal dependents or beneficiaries—serve as the core data sources for the machine

learning analytics engine. These inputs capture each client's measurable financial characteristics. Through data processing and predictive modeling, the system produces two key outcomes: the client's predicted poverty likelihood and the probability of becoming at-risk or delinquent. The outputs generated by the predictive model will be used as early warning signs that will enable proactive identification of financially vulnerable households and borrowers that require immediate attention.

The framework connects these predictive indicators to three main decision areas: selecting suitable loan products or services, guiding risk-mitigation actions based on borrower vulnerability and identifying and geo-mapping qualified scholarship recipients among clients' dependents. More accurate predictions of poverty status and borrower risk enable more precise, data-driven responses, such as targeted financial offerings, timely interventions to reduce default risk and fair allocation of scholarship resources in underserved areas. This framework positions machine learning as the link between raw client data and actionable decisions, demonstrating how analytics can improve client support, strengthen portfolio quality and enhance household welfare.

Figure 1

Conceptual Framework of the Study



Scope and Delimitations of the Study

Scope

This research assesses the contribution of CARD, Inc. through an analysis that integrates Progress out of Poverty Index (PPI) assessments, scholarship prioritization and predictive modeling for borrower risk identification. The primary focus is the development of a software system that utilizes these components to support decision-making in CARD, Inc. microfinance operations in Cavite, Laguna, Batangas, Quezon and Bicol Region. The analysis includes profiles of CARD, Inc. clients and their dependents to identify high-risk borrowers and provide data to maximize the distribution of financial and education services. The study population includes CARD, Inc. clients served in the past three years within the institution's

covered area. The objectives are to enhance operational efficiency, reduce loan defaults and ensure equitable, data-driven distribution of scholarships and loans services.

Delimitations

This study uses data exclusively from CARD, Inc. and does not include data from other microfinance institutions. Predictive modeling relies on selected variables such as client loan information and payment transactions, dependent details and PPI scores, which may not account for all external or unforeseen factors influencing borrower behavior. The scholarship prioritization framework follows CARD, Inc.'s criteria and may not apply to organizations with different eligibility requirements. The research addresses the development, design and evaluation of the software system, but does not assess its long-term implementation or operational outcomes beyond the study period.

Significance of the Research

This study will benefit the following:

Dependent and Borrowers. Borrowers and their dependents will benefit from a fairer and more consistent assessment process. With clearer and more accurate information, CARD, Inc. can better understand who truly needs support. This helps ensure that families facing financial hardship or educational challenges are recognized and given attention in a more transparent way.

Operation Staff. The system helps lighten the workload of operations staff by automating parts of the assessment process. Instead of manually reviewing large amounts of data, staff receive clearer insights that make it easier to identify at-risk clients and its dependents who may need support.

Policy Makers and Program Developers. For decision-makers within CARD, Inc., the study provides useful information that can guide the improvement of programs and services. With data-driven insights, they can design policies that respond better to the needs of clients and strengthen the CARD, Inc.'s outreach and support initiatives.

Researchers and Academics. The study adds to the growing field of data analytics in microfinance by showing how predictive modeling can support poverty assessment and risk evaluation. It can serve as a reference for future researchers interested in using technology to improve financial inclusion and social development programs.

Community and Society. When CARD, Inc. makes more informed decisions, the positive effects extend to the community. Borrowers may receive better-guided support, educational opportunities can reach those who need them most and communities can become more resilient. Over time, this contributes to reduced financial vulnerability and promotes a more inclusive and supportive environment for low-income households.

CHAPTER II

REVIEW OF LITERATURES

This chapter reviews relevant research, algorithms, theoretical frameworks and software applications for the proposed study. The review is organized by theme to highlight microfinance outcomes, program effectiveness and the use of data science in financial and social development.

The Role and Impact of Microfinance in Poverty Alleviation and Social Development

Microfinance is one of the key tools for advancing financial inclusion, poverty reduction and social development in developing countries. Microfinance institutions (MFIs) offer financial services to low-income populations excluded from the formal banking system, enabling entrepreneurship and improved livelihoods. Balaji (2024) notes that microfinance increases access to credit and savings while fostering entrepreneurship and economic empowerment. Ashraf et al. (2024) similarly find that microfinance initiatives significantly reduce poverty and support economic development in the Philippines.

Baniaga (2025) emphasizes that effective institutional practices and program design strengthen microfinance outcomes in Philippine universities and communities, particularly when aligned with community needs. Drean (2023) finds that microfinance programs also improve educational attainment and financial literacy, offering benefits beyond income growth. Chakraborty (2023) reports that

targeted microfinance programs help smallholder farmers increase household income and stabilize agricultural productivity.

Microfinance as a Tool for Economic Empowerment

Microfinance institutions (MFIs) are designed to extend financial services to low-income populations who are traditionally excluded from formal banking systems. According to Balaji (2024), this process promotes financial inclusion and encourages entrepreneurial activity. In support of this, Ashraf et al. (2024) assert that microfinance initiatives substantially contribute to both poverty alleviation and broader economic development.

Furthermore, microloans serve as the primary mechanism through which microfinance drives household income growth. For example, Almase (2021) found that microfinance interventions, particularly those combining loans with savings programs significantly improved household income and living standards in Lopez, Quezon. Likewise, Ashraf et al. (2024) reiterated the direct positive effects of microfinance programs on poverty reduction and community-level economic progress. Additional support for microfinance as an economic empowerment tool comes from Baniaga (2025), who emphasizes the role of institutional practices in enhancing program impact in Philippine universities and Chakraborty (2023), who reports similar financial benefits for small farmers.

Broader Social and Educational Outcomes

Beyond financial outcomes, microfinance programs support social and educational development. Acharya (2020) found that participation in Nepal

increased education and literacy among women and families. Drean (2023) similarly notes improved educational access and knowledge sharing in the Philippines.

More recently, research has emphasized the psychological impacts of microfinance. According to Gondraneos et al. (2025), community-based microfinance programs in the Philippines promote emotional resilience among low-income mothers. This finding suggests that microfinance groups provide not only material support but also socio-emotional support, which is crucial for long-term stability and empowerment. Fronda (2024) adds that microfinance interventions can enhance financial literacy and resilience among rural farmers, further demonstrating their multidimensional benefits.

MFI Service Offerings and Program Effectiveness in the Philippines

Effective microfinance implementation requires service offerings aligned with community needs. Sab et al. (2023) found that client awareness was highest for core services such as savings and microinsurance but notably lower for more specialized products like Regular Project Loans. These findings reveal the necessity for enhanced communication strategies to ensure that beneficiaries are fully informed about available services.

Additionally, Sab et al. (2023) emphasize that MFIs should prioritize essential services, including savings opportunities and microinsurance, to support long-term financial security. Almase (2021) also highlights the importance of savings-oriented programs in improving household well-being. Vicencio and

Villanueva (2024) note that service satisfaction directly affects the financial stability and participation of microfinance clients in Nueva Ecija.

Predictive Modeling and Financial Data Analysis in Microfinance

Modern microfinance institutions (MFIs) are increasingly adopting data-driven approaches to improve operational efficiency, strengthen risk management and support sustainable financial services. Machine Learning and predictive analytics have become essential tools for identifying repayment risks, optimizing loan allocation and improving decision-making processes. Abusharbeh (2023) emphasizes the importance of modeling internal institutional factors to predict Portfolio at Risk (PAR), a key indicator used to measure the likelihood of loan repayment problems. Through predictive modeling, MFIs can identify high-risk borrowers at an early stage and implement preventive interventions that help maintain portfolio quality and financial stability.

Similarly, Fajardo et al. (2024) demonstrated that machine learning models can accurately predict scholarship grant recipients, highlighting the effectiveness of predictive analytics in optimizing resource allocation within social programs. In the context of microfinance, similar predictive techniques can be applied to identify reliable borrowers, determine appropriate loan sizes and improve repayment performance. These applications support more informed lending decisions and reduce the possibility of default.

Several studies also emphasize the role of clustering and segmentation algorithms in financial analytics. Alam et al. (2021) found that clustering algorithms effectively segment clients according to repayment behavior, spending habits and demographic characteristics, allowing institutions to design targeted interventions and customized financial services. Oti et al. (2023) further discussed the effectiveness of K-Means and hierarchical clustering algorithms in uncovering hidden patterns within financial datasets and identifying meaningful client segments. These algorithms are particularly useful in microfinance because they enable institutions to group borrowers according to financial behavior and risk characteristics, thereby improving loan monitoring and client management strategies.

In addition to clustering methods, time-series forecasting techniques are valuable in evaluating repayment trends and monitoring portfolio health. Almeida et al. (2023) and Li and Law (2023) highlighted the importance of time-series analysis in forecasting financial performance and adapting to changing borrower behavior over time. Wauchope et al. (2023) similarly confirmed the effectiveness of time-series evaluation in monitoring the progress and sustainability of social programs. These forecasting approaches allow MFIs to anticipate repayment fluctuations, detect emerging risks and make timely operational adjustments.

To further improve managerial decision-making, Shubho et al. (2022) emphasized that real-time data visualization enhances organizational monitoring and operational efficiency among small and medium enterprises. This approach can support MFIs in monitoring loan releases, repayments and financial flows

through dashboards and analytics systems. Moreover, Sun et al. (2023) and Taylor and Brolos (2023) examined digital innovations in finance and e-commerce, demonstrating how technology-based systems streamline financial operations and expand financial access for underserved communities and smallholder farmers.

The evaluation of predictive models is equally important in ensuring the reliability and effectiveness of analytics systems used in MFIs. According to Žagar and Demšar (2022), predictive models must be evaluated using appropriate performance metrics to determine their accuracy and practical applicability. Vujović (2021) explained that classification algorithms are commonly assessed using evaluation metrics such as accuracy, precision, recall, specificity and Area Under the Curve (AUC). Accuracy measures the overall correctness of the model in predicting outcomes, while precision evaluates the proportion of correctly predicted positive cases among all predicted positive cases. Recall, also referred to as sensitivity, measures the ability of the model to correctly identify actual positive cases, whereas specificity determines the model's ability to correctly classify negative cases. AUC evaluates the model's capability to distinguish between positive and negative classifications across different threshold values.

Naidu et al. (2023) further emphasized that these evaluation metrics are essential in determining the predictive performance of machine learning algorithms, especially in financial and risk-related applications. In microfinance settings, these metrics are important in assessing how effectively predictive models can identify borrowers with high repayment risk while minimizing misclassification errors. The use of these evaluation measures ensures that the

developed predictive analytics system provides reliable, accurate and data-driven recommendations that can support lending decisions and portfolio management.

Similar Studies Combining Risk and Poverty Assessment, Scholarship Prioritization

Gutiérrez-Nieto and Serrano-Cinca (2023) show that evaluating microfinance institutions already involves multiple dimensions such as financial performance, outreach, governance, and social impact, but they also note that the impact of microcredit remains uncertain, which implies a need for more integrated and evidence-based assessment tools. Likewise, Anuar et al. (2024) map the microcredit research landscape and indicate that the field has expanded considerably, yet the literature is still largely organized around distinct themes rather than unified operational systems. Okesina (2025) further explains that microfinance research commonly examines poverty alleviation, financial inclusion, and borrower outcomes, but emphasizes persistent gaps, contradictions, and the tendency of studies to focus on borrower or institutional effects separately. On the risk side, Dako et al. (2024) discuss how micro-lending risk models and alternative credit scoring can improve inclusion and poverty reduction by generating more holistic borrower profiles, while Illangakoon (2024) finds that effective risk management is positively related to the sustainability of microfinance institutions. For scholarship support, Iligan et al. (2023) demonstrate that scholarship programs for microfinance clients' children can improve educational completion, employment prospects, and financial security, and they recommend strengthening selection

processes and grant design. However, based on these studies, no single existing study was found that fully combines borrower risk assessment, poverty assessment, and scholarship prioritization into one integrated system; instead, current literature discusses them as interconnected but standalone interventions, which highlights the research gap that this study seeks to address.

Synthesis

The reviewed literature establishes that microfinance plays a significant role in poverty alleviation, financial inclusion, and social development by improving access to credit, savings, education, and financial literacy among low-income communities. Studies by Balaji (2024), Ashraf et al. (2024), and Almase (2021) consistently emphasize the positive economic effects of microfinance programs, while Acharya (2020), Drean (2023), and Gondraneos et al. (2025) highlight their broader social and psychological benefits. In the Philippine context, the reviewed studies further reveal that effective institutional practices, client-centered services, and financial literacy initiatives strengthen the impact of microfinance programs. These findings are relevant to the present study because they demonstrate the importance of developing systems that not only support financial assistance but also improve decision-making, borrower monitoring, and social support mechanisms within microfinance institutions.

The literature also shows that predictive analytics, machine learning, clustering algorithms, and time-series forecasting are increasingly being utilized to enhance financial analysis and risk management in microfinance and related social

programs. Studies by Abusharbeh (2023), Fajardo et al. (2024), Alam et al. (2021), and Oti et al. (2023) demonstrate that predictive models can improve borrower assessment, client segmentation, and resource allocation, while researchers such as Almeida et al. (2023) and Li and Law (2023) emphasize the usefulness of forecasting techniques in monitoring financial sustainability and repayment behavior. Furthermore, studies on model evaluation stress the importance of accuracy, precision, recall, specificity, and AUC in ensuring reliable predictive performance. These studies support the present research by providing the theoretical and technical foundation for integrating predictive analytics into a microfinance decision-support system.

Despite the growing body of literature on microfinance, risk assessment, and scholarship assistance, the reviewed studies reveal a significant research gap. Existing studies generally focus on borrower risk evaluation, poverty reduction, scholarship prioritization, or institutional performance as separate areas of investigation. Although studies such as Dako et al. (2024), Illangakoon (2024), and Iligan et al. (2023) discuss interconnected aspects of financial risk management and educational support, no existing study was found to integrate borrower risk assessment, poverty assessment, and scholarship prioritization into a single analytics-driven system. Therefore, the present study aims to address this gap by developing an integrated predictive analytics system that combines financial risk evaluation and scholarship prioritization to support more effective, data-driven, and socially responsive decision-making in microfinance institutions.

Overall, the literature confirms that microfinance significantly contributes to poverty reduction, financial inclusion and social empowerment, while predictive analytics and digital technologies strengthen institutional performance and sustainability. However, current approaches remain fragmented and fail to integrate poverty assessment, borrower risk analysis and scholarship prioritization into a unified system. Consequently, there is a need for a comprehensive and data-driven framework that combines these components to support more effective and sustainable microfinance interventions. The present study addresses this gap by proposing an integrated system for CARD, Inc. that utilizes predictive modeling and poverty assessment to improve scholarship allocation, borrower evaluation and institutional sustainability.

CHAPTER III

RESEARCH METHODOLOGY

This chapter presents the research methodology, describing the approaches and procedures used to collect and analyze data.

Research Design

This study uses experimental research designs to assess borrower risk, evaluate client poverty levels through the Progress out of Poverty Index (PPI) and identify scholarship priorities for CARD, Inc. clients. Client's information, loan histories, repayment behavior, PPI results and dependent information are analyzed to support model development. These data provide a structured view of client conditions, enabling the system to generate predictions and recommendations that improve the accuracy and efficiency of borrower assessment and scholarship prioritization.

The experimental design is applied to test and compare different machine learning algorithms, such as Logistic Regression, Random Forest, Decision Tree, and Gradient Boosting, to identify which model achieve the highest predictive accuracy for borrower risk and PPI estimation. Following standard practices in computer science research, these algorithms are evaluated under controlled conditions using metrics like accuracy, precision, recall, F1-score and error rates (Sculley & Pasanek, 2011). This process helps ensure the chosen model is reliable and suitable for decision-making in CARD, Inc.

By using both descriptive and experimental model testing, the study builds a strong base for accurate assessments and better decision-making in microfinance operations.

Locale of the Study

The study was conducted at CARD, Inc. (A Microfinance NGO), a microfinance institution based in San Pablo City, Laguna, Philippines. As an organization that works closely with low-income households in both rural and semi-urban communities nationwide, CARD, Inc. provides an ideal environment for testing an analytics-driven system for borrower risk assessment, poverty evaluation and scholarship prioritization. Its wide and diverse client base offers rich opportunities to analyze financial behavior and understand varying levels of household vulnerability.

The study used data from CARD, Inc.'s internal system, such as client details, loan and repayment records, PPI results and beneficiary lists. These datasets capture real client behavior and situations, making them useful for building models to predict risk and automated assessment of poverty. Using this operational data helps the system address real challenges, such as identifying high-risk borrowers early, predicting poverty and making decisions that are consistent across branches.

Applied Concepts and Techniques

This study is grounded in data science, machine learning and predictive analytics to identify at-risk borrowers, assess clients' progress out of poverty and prioritize scholarship support. Its main theoretical foundation is supervised

machine learning, which relies on labeled historical data to generate predictions about future outcomes.

To understand borrower behavior and predict risk levels, the study applies algorithms such as Random Forest, Logistic Regression, and Gradient Boosting. These models help uncover meaningful patterns in repayment activities and financial histories. K-means clustering and spatial regression are also used to group clients and analyze geographic trends.

Classification techniques categorize borrowers into Low, Moderate, or High Risk, while data mining helps reveal hidden patterns that may indicate potential repayment difficulties. Predictive analytics further supports these processes by using statistical modeling to estimate the likelihood of default or financial strain.

Machine learning is also applied to predict and assess clients' PPI scores, allowing the system to estimate a household's likelihood of progressing out of poverty. This data-driven method provides a more proactive way of monitoring client well-being and identifying vulnerable or improving households.

Additionally, the study incorporates geo-mapping to guide scholarship prioritization. By examining clients' locations, branch coverage, dependent's grade school level and the number of dependents, the system identifies areas with higher need and supports a more equitable distribution of scholarships. Visualizing spatial patterns in poverty levels, population distribution and dependent counts allows CARD, Inc. to make more informed and targeted decisions.

Together, these methods create a theoretical and methodological foundation for developing a practical, predictive analytics and geo-mapping system aligned to the needs of CARD, Inc.

Algorithm Analysis

This study examines a multi-faceted machine learning approach designed to strengthen data-driven decision-making in CARD, Inc.'s microfinance and poverty-alleviation programs. The integrated system's algorithms were evaluated for their performance, efficiency, and suitability in predicting both at-risk borrowers and client poverty levels through the Progress Out of Poverty Index (PPI), while also supporting scholarship prioritization. Each model was assessed using standard classification metrics and measures of computational complexity to ensure that the system can efficiently process CARD, Inc.'s large client database. These evaluation standards help ensure that the predictive and decision-making components operate reliably in real-world microfinance environments.

Supervised learning models such as Logistic Regression, Random Forest, and Gradient Boosting were applied to predict borrower risk and classify client poverty levels based on historical financial behavior, repayment patterns, demographic characteristics, and PPI-related indicators. Their performance was measured using AUC, accuracy, precision, recall, specificity and F1-score to assess predictive accuracy and overall reliability (Yacouby & Axman, 2020).

Accuracy, which measures the proportion of correctly predicted borrowers relative to the total number of cases:

ACCURACY = TP + TN / TP + FP + TN + FN

Precision, which measures the proportion of borrowers identified as at-risk who are truly at-risk:

PRECISION = TP / TP + FP

Recall, or sensitivity, which determines how many actual at-risk borrowers were correctly detected:

RECALL = TP / TP + FN

F1 Score, the harmonic mean of precision and recall, which provides a balanced evaluation especially useful when the classes (risk vs. non-risk) are imbalanced:

F1 = 2 × (Precision × Recall) / (Precision + Recall)

Specificity, which measures how accurately the model identifies low-risk borrowers:

SPECIFICITY = TN / TN + FP

In addition, the Receiver Operating Characteristic (ROC) curve and its Area Under the Curve (AUC) were used to assess the model's overall ability to distinguish between at-risk and non-risk clients. A higher AUC indicates better discriminative performance, which is essential for reliable borrower risk prediction.

In the context of scholarship prioritization, the study applies a rule-based approach to group and assess clients. The system organizes clients based on their assigned area or unit office, reflecting the existing operational structure of CARD, Inc. This allows for a more practical and manageable way of reviewing beneficiaries within specific geographic assignments.

The prioritization framework is based on key indicators, particularly the client's Progress Out of Poverty Index (PPI) score and risk classification. Clients with lower PPI scores and lower risk levels are prioritized, as they represent households that are economically vulnerable yet relatively stable. In addition, priority is further influenced by the number of dependents and their educational level, with higher priority given to clients who have more dependents and those with children in High School, Senior High School, and College. This approach ensures a more practical, transparent, and organization-aligned method of distributing limited educational resources while maintaining a data-driven basis for decision-making.

These analyses demonstrate how machine learning can be integrated into a predictive and prescriptive analytics system that advances CARD, Inc.'s mission of delivering inclusive, targeted and effective services.

Data Collection Methods

This study utilized data extracted from CARD, Inc.'s core system, consisting of structured quantitative records such as client profiles, loan and repayment histories, PPI (Progress out of Poverty Index) score and beneficiary information. These datasets contain essential operational indicators that support predicting borrower risk, estimating poverty likelihood and prioritizing scholarship beneficiaries. Data extraction will be coordinated with CARD, Inc.'s Information Technology team, who will prepare the necessary datasets for a defined period to ensure data privacy and compliance with institutional protocols. Prior to model

development, the data will undergo cleaning and validation to ensure accuracy and reliability for machine learning and prescriptive analytics.

Data collection will involve retrieving CSV files directly from the core system. These datasets will serve as the foundation for building predictive models and generating data-driven recommendations. Table 1 presents an overview of the primary datasets and attributes used in the study, including their descriptions and sample values.

Table 1
Dataset Attributes

Attribute Name	Description	Sample Value
Area	Geographic area where the clients belong	Laguna 1, Laguna 2, Batangas 1
Unit	Operational unit under the area; sub-branch grouping.	Bay, San Pablo, Paete,
Center	Center name or center group where the client is assigned.	Magsaysay 1, Balayhangin, San Juan 1
CID	Client Identification Number; a unique code for each borrower.	10001212, 10100221
Client Name	Full name of the client	Maria Dela Cruz
Length of Membership	How long the client has been a member of CARD, Inc.	ACC-2024-9987
Age	Age of the client	24,31,45
Civil Status	Marital status of the client	Married, Single, Widow
Gender	Gender of the client	Female, Male
Business Type	Type of business owned or managed	Wholesale and Retail Trade
Employment	Client's job or work status	Employed, Self-Employed
Barangay	Smallest local area/community	Balayhangin
Municipality/City	Town or city	Bay
Province	Province location	Laguna
Household Monthly Income	Total income of the household per month	5,000-15,000
Household Monthly Expenses	Total monthly spending of the household	5,000-10,000
No. of Children	Number of children	1, 2, 5
No. Legal Dependent	Number of dependents supported by the client	0, 5, 6, 7
Name of Legal Dependent/Beneficiary	Name of dependent	Joseph Dela Cruz
Age	Age of dependent	2, 10, 16
Birthday	Birthdate of dependent	22-Jan-00

Grade School	Education level of dependent	Grade 5, Grade 10
Loan Cycle	Number of times the client has taken a loan	5, 8, 12
Loan Account Number	Unique loan account ID/number	4018-0000-00148926
Loan Type	Type/category of loan	Sikap 1, Home Improvement Loan
Loan Amount	Loan amount granted/released	12,000, 15,000
Loan Balance	Remaining unpaid loan	9772.04
Weekly Due	Weekly payment due/amount	290
Loan Term	Duration of the loan to be paid	23
Term Paid	Number of terms paid	1, 5, 10
Days Arrears/Unpaid	Number of days payment is overdue	14, 32, 56
Late Payment Last 8 Weeks	Count of late payments in the past 8 weeks	7,11,15
Savings Balance	Current savings balance	8296.35
Savings Balance last Month	Savings in previous month	1548.31
Savings Balance current Month	Savings this month	8296.35
PPI Score	Poverty score indicating likelihood of being below poverty line	35, 46, 52
PPI Question 1	Scores from survey questions used to compute PPI	0,4,9, 15, 20, 26
PPI Question 2	Scores from survey questions used to compute PPI	0,2,4
PPI Question 3	Scores from survey questions used to compute PPI	0,3,6,11
PPI Question 4	Scores from survey questions used to compute PPI	0,5
PPI Question 5	Scores from survey questions used to compute PPI	0,4
PPI Question 6	Scores from survey questions used to compute PPI	0,2
PPI Question 7	Scores from survey questions used to compute PPI	0,7
PPI Question 8	Scores from survey questions used to compute PPI	0,10
PPI Question 9	Scores from survey questions used to compute PPI	0,6,21
PPI Question 10	Scores from survey questions used to compute PPI	0,10
Payment Compliance Ratio	How consistently the client pays on time	0.7, 0.002
Savings Trend %	Increase or decrease in savings over time	0.5,0.2
Transaction Trend %	Change in transaction activity over time	0.8,0.9
Exposure Ratio	Level of financial risk based on loan vs capacity	0.4,1.1

At Risk Client Scoring. The computation of the at-risk client score is based on a set of selected socio-economic and behavioral factors that reflect the repayment capacity and financial stability of borrowers within CARD, Inc. These factors include client age, payment behavior (e.g., late payments in the last eight weeks and days past due), length of membership, loan cycle, and compliance and repayment progress ratios. In addition, financial indicators such as declared household monthly income, savings trend, and transaction trend are incorporated to capture the client's financial discipline and cash flow patterns. The inclusion of the Progress out of Poverty Index (PPI) further strengthens the assessment by providing a standardized measure of the client's poverty likelihood, which is relevant in identifying vulnerable borrowers. Each factor is assigned a corresponding maximum score, contributing to an aggregated risk score that serves as the basis for classifying clients according to their level of risk.

The scoring mechanism is designed to provide a structured and quantifiable approach to risk assessment by assigning weights to each factor based on its perceived importance in predicting repayment behavior. Behavioral indicators, such as late payments and repayment progress ratio, are given emphasis as they directly reflect the client's credit discipline. Meanwhile, demographic and socio-economic variables, such as number of children and income level, offer supporting context that may influence a client's financial obligations and capacity to repay. By combining these variables, the model produces a composite score that enables monitoring officers to systematically identify clients who may require closer monitoring or intervention.

This scoring framework is specifically developed for the operational context of CARD, Inc. and is tailored to its data structure, client profile, and institutional policies. As such, the applicability of this scoring system is limited to CARD, Inc., and the selected factors and their corresponding weights may not be generalizable to other organizations or financial institutions. Furthermore, while the model incorporates key indicators relevant to risk assessment, there may be other external or unobserved factors, such as sudden economic changes, personal emergencies, or informal income sources, that are not captured in the current scoring system. Therefore, the results should be interpreted as a decision-support tool rather than a definitive measure of client risk.

Progress Out of Poverty Index Scoring. The scoring of the Progress Out of Poverty Index (PPI) is used to assess the poverty likelihood of clients based on selected household and socio-economic indicators captured through the standard PPI questionnaire. Each client is assigned a corresponding score derived from these indicators, which are then aggregated to produce a total PPI score. This score serves as the basis for segmenting clients into distinct categories that reflect their economic standing. In this study, the classification follows the internal standard used by CARD, Inc., wherein clients with a total score of 34 and below are categorized under the Pagtalubo segment, while those with a score of 35 and above are classified as Regular clients.

This classification supports CARD, Inc.'s targeted interventions by providing a structured basis for identifying vulnerable clients who may require additional assistance. The integration of PPI scoring into the system enables monitoring

officers to make consistent and data-driven assessments aligned with institutional practices. However, it is important to note that this segmentation is based solely on the indicators included in the PPI questionnaire. Other potential factors influencing a client's economic condition such as informal income sources, sudden financial shocks, or unrecorded household circumstances are not captured and therefore not considered in the classification. The PPI-based segmentation is interpreted as a standardized internal tool of CARD, Inc., rather than a comprehensive measure of a client's overall economic status.

Figure 2

Progress Out of Poverty Index Scorecard

PROGRESS OUT OF POVERTY INDEX (PPI) scorecard for the Philippines is based on the 2004 Annual Poverty Indicators Survey (APIS) V.6142011

Indicator	Pakibilugan ang mga sagot ng kinapanayam/interviewee (Response)						Score
1. Ilan po sa inyong pamilya ang nasa **edad 14 pababa?**	Lima o higit pa	Apat	Tatlo	Dalawa	Isa	Wala	
	0	4	9	15	20	26	
2. Lahat po ba ng batang may **edad na 6-14** sa inyong pamilya ay **pumasok sa eskwelahan?**	Hindi		Oo		Walang bata na may edad na 6-14		
	0		2		4		
3. Ano po ang **pinakamataas na antas** ng pag-aaral ang **natapos ng babaeng puno ng pamilya/asawang babae?**	Nagtapos sa elementarya o mas mababa pa	Una hanggang ika-apat na taon sa hayskul	Nagtapos sa hayskul	Unang taon sa kolehiyo o higit pa, o walang babaeng puno ng pamilya			
	0	3	6	11			
4. Mayroon po bang miyembro ng inyong pamilya ang tumatanggap ng **regular na sweldo?**	Wala		Meron				
	0		5				
5.Anong materyales po gawa ang inyong **dingding?**	Magaan na materyales (cogon, nipa o sawali, kawayan, anahaw)		Matibay na materyales (bakal, *aluminum*, tile, konkreto, tisa, bato, kahoy, asbestos)				
	0		4				
6. Anong materyales po gawa ang inyong **bubong?**	Magaan na materyales (gamit na makeshift, cogon, nipa, o anahaw)		Matibay na materyales (yero, *aluminum*, tile, konkreto, tisa, bato o asbestos)				
	0		2				
7.Ano po ang **uri ng palikuran o toilet** mayroon ang inyong pamilya?	Wala, open pit, close pit, o iba pa		De buhos (water sealed)				
	0		7				
8. Mayroon po ba kayong **gumaganang refrigirator? Sa inyo po ba ito?**	Wala		Meron				
	0		10				
9. **Ilan** po ang **pag-aari ninyong telebisyon? Gumagana** po ba ang mga ito?	Wala		Isa		Dalawa o higit pa		
	0		6		21		
10. Mayroon po ba kayong **gumaganang washing machine? Sa inyo po ba ito?**	Wala		Meron				
	0		10				
					TOTAL POVERTY SCORE		

Scholarship Allocation Prioritization. The prioritization of scholarship allocation is based on the integration of client-dependent information, computed risk score, and Progress Out of Poverty Index (PPI) score within the system. The process begins by capturing relevant details of the client's dependents, including their number and educational level, which are then linked to the corresponding client profile. These data points are evaluated alongside the client's risk classification and PPI score to determine eligibility and level of priority for scholarship support. This approach ensures that both the financial vulnerability and repayment condition of the client, as well as the educational needs of their dependents, are considered in identifying beneficiaries.

The prioritization mechanism follows a need-based logic, wherein clients with lower risk scores and lower PPI scores are given higher priority for scholarship allocation. A lower PPI score indicates a higher likelihood of poverty, while a lower risk score indicates that the client is financially stable enough to sustain support, making them suitable candidates for scholarship assistance. In addition, clients with a higher number of dependents are given increased priority, as this reflects greater financial responsibility and potential need. Special consideration is also given to dependents at the grade school level, dependent in High School, Senior High School and College are given higher prioritization.

This method aligns with the objective of providing targeted and equitable support to the most deserving beneficiaries. However, the prioritization is limited to the variables captured within the system, and other external factors that may affect a dependent's educational needs are not included in the allocation process.

Data Model Generation

The data model generation process follows the CRISP-DM methodology, beginning with a structured review of CARD, Inc.'s operational datasets and progressing toward analytical models for borrower risk prediction, poverty assessment and scholarship prioritization. Data from various sources, including loan histories, client demographics available within the CARD, Inc. core system, PPI survey results and dependent information, were generated to form a analytical dataset. Relationships among variables were examined and mapped to convert financial behavior, household vulnerability and demographic attributes into meaningful predictive features.

This resulting data structure served as the foundation for supervised learning models that produce three core outputs: borrower risk classifications, predicted PPI values indicating poverty likelihood and scholarship prioritization indicators based on dependent information and geographic clustering. The design of the model supports interpretability, facilitates validation and allows for smooth integration into the prescriptive analytics system prototype.

Data Collection and Cleaning

Data collection will follow the CRISP-DM Data Understanding phase, which involved extracting structured datasets from CARD, Inc.'s core system. These datasets will include borrower profiles, loan disbursements, repayment transactions, PPI scores and dependent information. Because these records reflect real

operational activities, they provide a dependable basis for models that align with CARD, Inc.'s actual decision-making environment.

During the Data Preparation phase, data cleaning will be performed to ensure accuracy and readiness for modeling. This process involved:

- Removing duplicate records resulting from repeated transactions,
- Correcting inconsistent labels, such as mismatched loan types or category entries,
- Standardizing formats for dates, numerical fields and categorical variables.

These steps will produce a consistent and high-quality dataset, suitable for developing reliable predictive and prescriptive analytics models.

Feature Engineering

Feature engineering plays a critical role in transforming raw operational data into meaningful predictors, in line with the CRISP-DM Data Preparation phase. A range of financial, demographic and socio-economic indicators will be constructed to enhance the accuracy and relevance of the models.

Key engineered features included:

- Repayment behavior metrics, such as days late, delinquency patterns and frequency of missed payments.
- PPI-based vulnerability indicators drawn from asset deprivation and household characteristics.

- Dependent-related variables, such as the number of dependents and their distribution across education levels.
- Geo-spatial attributes, including client density per branch and cluster membership for scholarship prioritization.

Feature scaling, through standardization or normalization, was applied to maintain consistency across model inputs. Irrelevant or highly correlated variables were removed based on correlation analysis, multicollinearity checks and model-based feature importance. This process ensured that only the most predictive and interpretable features were retained for modeling.

Model Building

Model building follows the CRISP-DM Modeling phase and develops predictive models for three key analytical objectives:

- Borrower Risk Prediction – Classification models, including Logistic Regression, Random Forest and Gradient Boosting, were selected for their proven effectiveness in credit scoring and ability to handle nonlinear relationships.
- PPI Score Assessment – Regression models, such as Random Forest Regressor, Linear Regression and Gradient Boosting, were trained to estimate current and future PPI values. This approach supports early identification of households with rising vulnerability.
- Scholarship Prioritization Framework – Outputs from the borrower's risk and PPI score, number of dependents and grade school level where combined,

rule-based approached and operational to identify high-need geographic groups with many dependents and low PPI scores.

Model parameters were optimized through cross-validation to reduce the risk of overfitting and ensure that the models generalized well to unseen data. Final model selections were based not only on their accuracy, but also on their interpretability and suitability for CARD, Inc.'s operational requirements.

Model Evaluation

Model evaluation will follow the CRISP-DM Evaluation phase to ensure that each predictive component performed reliably before being integrated into the system. For borrower risk classification, models such as Logistic Regression, Random Forest and Gradient Boosting were assessed using accuracy, precision, recall, specificity, F1-score and AUC. Together, these metrics provided a well-rounded understanding of how effectively each model distinguished between high- and low-risk borrowers, especially in the presence of dataset imbalance. Insights from the confusion matrix helped identify misclassification trends and guided further feature refinement and model tuning.

For Progress Out of Poverty Index (PPI) prediction, the same evaluation metrics such as accuracy, precision, recall, specificity, F1-score, and AUC were applied to ensure consistency across predictive components. These metrics were used to assess how effectively the models classify clients into poverty categories and distinguish between different levels of vulnerability. This unified evaluation approach allows for a more direct comparison between borrower risk and poverty

classification models, ensuring that both components meet the required standards of accuracy, reliability, and practical applicability within the system.

The scholarship prioritization component is not based on machine learning techniques such as clustering or spatial analytics. Instead, it follows a rule-based approach that organizes clients based on the geo-location of their assigned offices, which are already tagged within the system. This geographic grouping allows monitoring officers to easily review and manage potential beneficiaries within their respective areas, ensuring alignment with existing operational structures.

Prioritization is determined using key indicators derived from the system, including the number of dependents, presence of dependents at the grade school level, Progress Out of Poverty Index (PPI) score, and at-risk classification. Clients with lower PPI scores and lower risk classifications are given higher priority, as they represent households that are economically vulnerable but relatively stable. Additional priority is given to clients with a higher number of dependents and children in High School, Senior High School and College.

Only models that demonstrated a strong balance of accuracy, interpretability and practicality for CARD, Inc.’s operations were selected for system integration.

Model Deployment

Following the CRISP-DM Deployment phase, the selected models will be integrated into the system prototype to generate real-time borrower risk scores, poverty likelihood predictions and scholarship prioritization outputs. API endpoints

will be developed so that the web application could send input data, run model inference and display predictions directly on user dashboards.

The system presents:

- Borrower risk levels (Low, Moderate, or High),

Table 2
At-Risk Scoring

SN	At-Risk Score	Borrower Risk Level
1	1-30	Low
2	31-80	Moderate
3	80-90	High

- Predicted PPI scores indicating poverty risk,

Table 3
Progress out of Poverty Index Scoring

SN	PPI Score	Classification
1	1-34	Pagtalubo
2	35-100	Regular

- Scholarship priority clusters highlighting underserved communities.

Table 4
Scholarship Rule-Based Factors

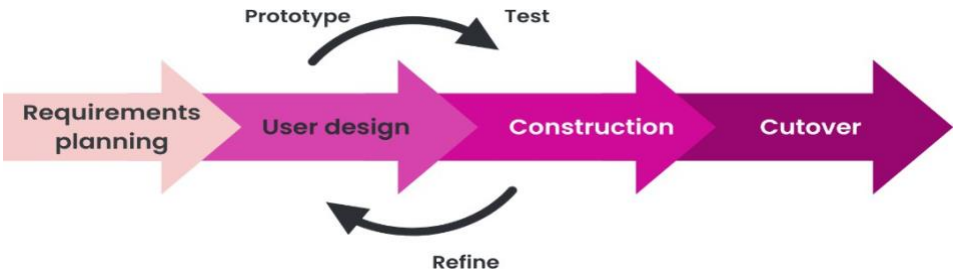
PPI		At-Risk Level		Grade Level		No. of Dependent	
Legend	Score	Legend	Score	Legend	Level	Legend	Count
Low	1-34	Low	1-30	Low	Grade 1-6	Low	1
High	35-100	Moderate	31-80	Medium	Grade 7-10	Medium	2-4
		High	81-100	High	SHS/College	High	5-10

Although full real-time integration with CARD, Inc.'s operational systems is beyond the scope of this academic project, the prototype supports batch data processing for periodic updates of client risk and poverty assessments. The deployment design also allows for model retraining, ensuring that predictive accuracy improves over time as new borrower and PPI data become available.

System Development Methodology

The system will be developed using the Rapid Application Development (RAD) methodology, an iterative and user-centered approach that emphasizes rapid prototyping and continuous feedback. RAD was chosen because it supports the seamless integration of machine learning outputs into a practical system that CARD, Inc. personnel could review and refine throughout the development process. The methodology is composed of four phases, Requirements Planning, User Design, Construction and Cutover, each contributing to a faster and more collaborative development cycle.

Figure 3
Rapid Application Development



Note: Adapted from https://solaceinfotech.com/blog/what-is-rapid-application-development-rad-why-when-should-you-use-it

Requirements Planning

During the Requirements Planning phase, the essential system components needed to support borrower risk classification, PPI assessment and scholarship prioritization were identified. Through consultations with CARD, Inc. monitoring officers and staff, both functional requirements, such as dataset uploads, predictive output generation and visualization of results and non-

functional requirements, including usability, accuracy and reliability, were established. This phase ensured that the system aligned with existing organizational workflows and could effectively support evidence-based decision-making.

User Design

In the User Design phase, initial prototypes of the system's screens and workflows will be created, including early layouts for data upload modules, borrower risk displays, PPI summaries and scholarship prioritization interfaces. CARD, Inc. users reviewed these prototypes and provided ongoing feedback, leading to improvements in navigation, clarity and overall workflow design. This collaborative process helped ensure that the system's interface was intuitive, user-friendly and closely aligned with real operational practices.

Construction

During the Construction phase, the system's frontend interface, backend engine and integrated machine learning models will be developed. The backend will handle data preprocessing and execute the predictive models, while APIs facilitated communication between the user interface and the analytics engine. The frontend will present model outputs in a clear and user-friendly format. Development will follow iterative cycles, allowing rapid revisions based on user feedback and ensuring that all core features, such as borrower risk scoring, PPI prediction and geo-mapped scholarship prioritization, will be fully implemented and functioning as intended.

Cutover

The Cutover phase will involve final testing, user acceptance testing and deployment of the operational prototype. CARD, Inc. personnel will conduct user acceptance testing to verify the system's functionality, clarity and ease of use. Based on their feedback, minor refinements will be made and the prototype will be prepared for demonstration and limited-use deployment. Although a full institutional rollout is beyond the scope of this academic study, the Cutover phase will confirm that the system effectively performs its intended functions of risk prediction, poverty assessment and scholarship prioritization in alignment with CARD, Inc.'s operational needs.

The Cutover phase will includ final testing, user acceptance evaluation and deployment of the operational prototype. CARD, Inc. personnel will validate system functionality, clarity and usability through user acceptance testing. Minor refinements will be made based on feedback and the final prototype will be prepared for demonstration and limited-use deployment. While a full institutional rollout is outside the scope of this research, the Cutover phase will confirm that the system meets its intended functions of risk prediction, poverty assessment and scholarship prioritization in line with CARD, Inc.'s operational requirements.

Software Tools Used

This study employed several software tools for data preprocessing, machine learning model development and the development of a web-based

predictive analytics system. Tools will be chosen for their reliability, integration and ability to manage CARD, Inc.'s operational datasets.

Scikit-learn Library

Scikit-learn will be used to develop and evaluate machine learning models for borrower risk classification and PPI score prediction. Its built-in algorithms and utilities supported preprocessing, model training and performance evaluation, allowing for efficient experimentation and refinement.

PostgreSQL

PostgreSQL will be the primary database for storing structured datasets, including borrower profiles, loan histories, PPI results and geospatial information. Its robustness, scalability and analytical capabilities made it well-suited for managing the study's large institutional datasets.

pgAdmin

pgAdmin will be used to manage the PostgreSQL database through a graphical interface. It will support data inspection, query execution and maintenance of tables for risk prediction, PPI scoring and scholarship prioritization.

Python Programming Language

Python is the core language used for data cleaning, preprocessing, exploratory analysis and machine learning model development. Its extensive ecosystem enabled seamless integration of predictive models with the web-based system's backend components and facilitated the automation of analytical workflows.

Visual Studio Code

Visual Studio Code is the primary development environment for backend scripts, API endpoints and other system components. Its debugging tools and plugins streamlined thedevelopment and testing of the predictive analytics application.

Web Development Tools (HTML, CSS, JavaScript)

HTML, CSS and JavaScript will be used to develop the system's user interface. These technologies will enable CARD, Inc. staff to have a responsive, user-friendly experience. Browser-based testing, primarily in Google Chrome, ensured compatibility and usability across common devices.

Hardware Requirements

The hardware will be used in developing and testing the system reflected typical operational conditions within CARD, Inc., ensuring that the predictive analytics system remains accessible even on modest workstations.

The desktop computer will be used during development provided stable performance for data processing, model training and system integration. Its specifications supported the execution of Python scripts, Scikit-learn models and web technologies without significant delays, ensuring consistency throughout the development cycle.

Table 5
Minimum Hardware Requirements

Component	**Specification**
Processor	Intel® Core™ i3-4030 CPU @ 3.30GHz
Memory	2.00 GB
Hard Disk Drive	10 GB
System Type	32-bit Operating System

System Architecture

The system architecture uses a modular client–server design that brings together data sources, machine learning processes, backend operations and a user-friendly web interface. Each component works in coordination to deliver accurate borrower risk predictions, PPI-based poverty assessments and scholarship prioritization insights for CARD, Inc.

The **Data Source Layer** serves as the foundation of the system. It gathers essential datasets from CARD, Inc.'s core banking system, including loan histories, repayment transactions, PPI survey scores, dependent information and demographic details. These datasets supply the attributes needed for machine learning models to produce borrower risk classifications, estimate poverty likelihood and identify scholarship-priority areas using geo-mapping. The layer supports both numerical and categorical data formats, ensuring seamless integration with preprocessing and analytics components.

The **Data Processing and Analytics Engine** serves as the core computational layer of the system. It receives client, loan and PPI data from the database, performs preprocessing tasks and executes trained machine learning models. From this process, it generates borrower risk classifications, predicted PPI values and scholarship priority indicators. Built for efficiency and scalability, the engine ensures that results are produced quickly and consistently, supporting both real-time and batch decision-making workflows.

The **Application Server** facilitates communication between the analytics engine and the user interface. Developed using Python-based frameworks, it handles API requests, manages user sessions and oversees data flow between system layers. When users submit borrower information, the server forwards the data to the analytics engine, retrieves the model outputs and delivers the results back to the interface. Its RESTful structure promotes modularity, system stability and compatibility with potential future integrations within CARD, Inc.

The **User Interface Module** is a web-based platform built using HTML, CSS, JavaScript and Bootstrap. It provides intuitive forms for entering borrower information and presents predictive results in clear, easy-to-interpret formats. Through this interface, users can access borrower risk classifications, PPI assessments and scholarship prioritization outputs. The design prioritizes clarity, usability and accessibility so that CARD, Inc. monitoring officers can comfortably navigate the system and interpret insights during both fieldwork and office assessments.

The system architecture and sequence diagram integrates data collection, machine learning analysis and user interaction within a unified framework. Its modular design supports accuracy, scalability and operational relevance, aligning with the study's goal to improve borrower risk identification, poverty assessment and decision-making at CARD, Inc.

Figure 4
Sequence Diagram

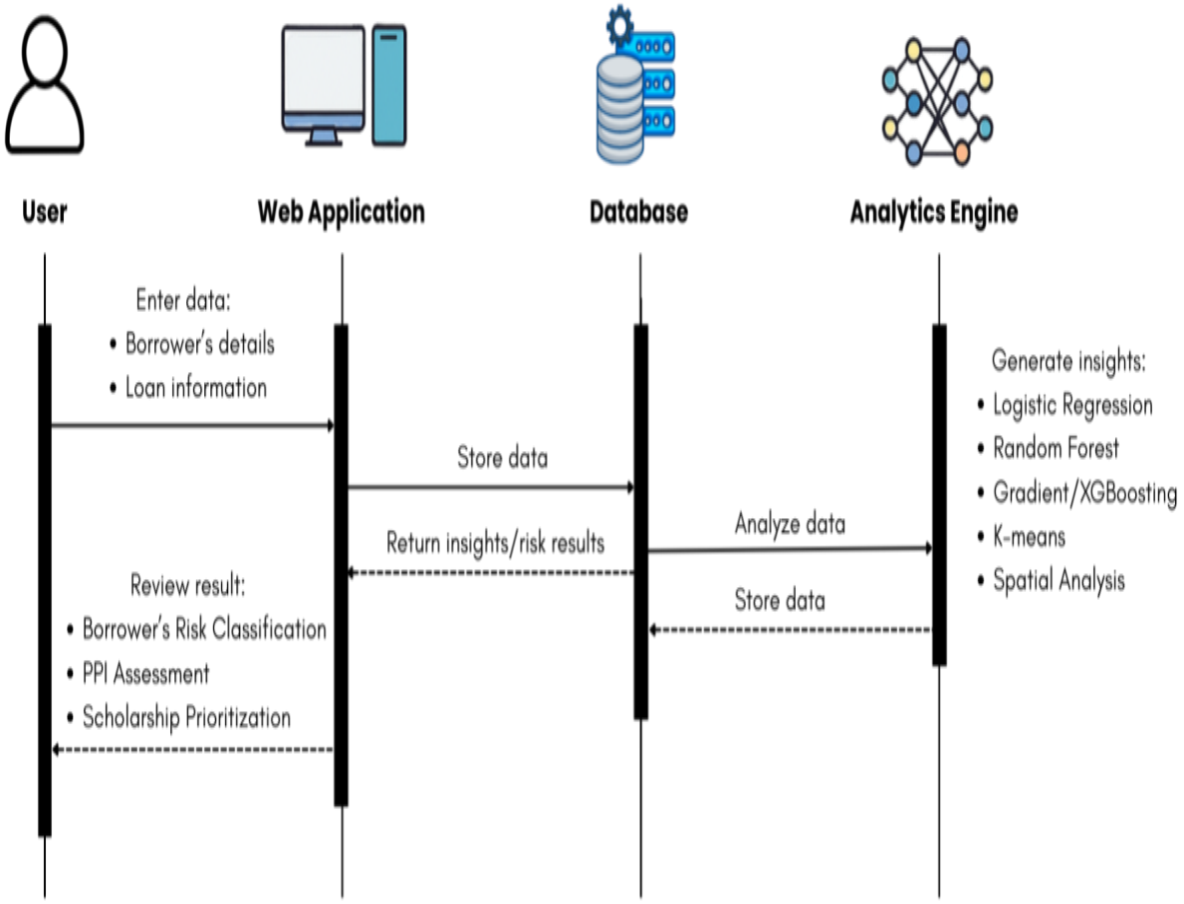


Figure 4 shows the chronological interaction between the user and the major components of the system. The process starts when the user enters borrower and loan information through the web application. The application communicates with the database to store and retrieve the necessary information. The retrieved data are then analyzed by the analytics engine using the selected machine learning and analytical techniques. After processing, the system returns the generated insights to the web application, where the user can review the borrower risk classification, PPI assessment, and scholarship prioritization results.

Figure 5
System Architecture

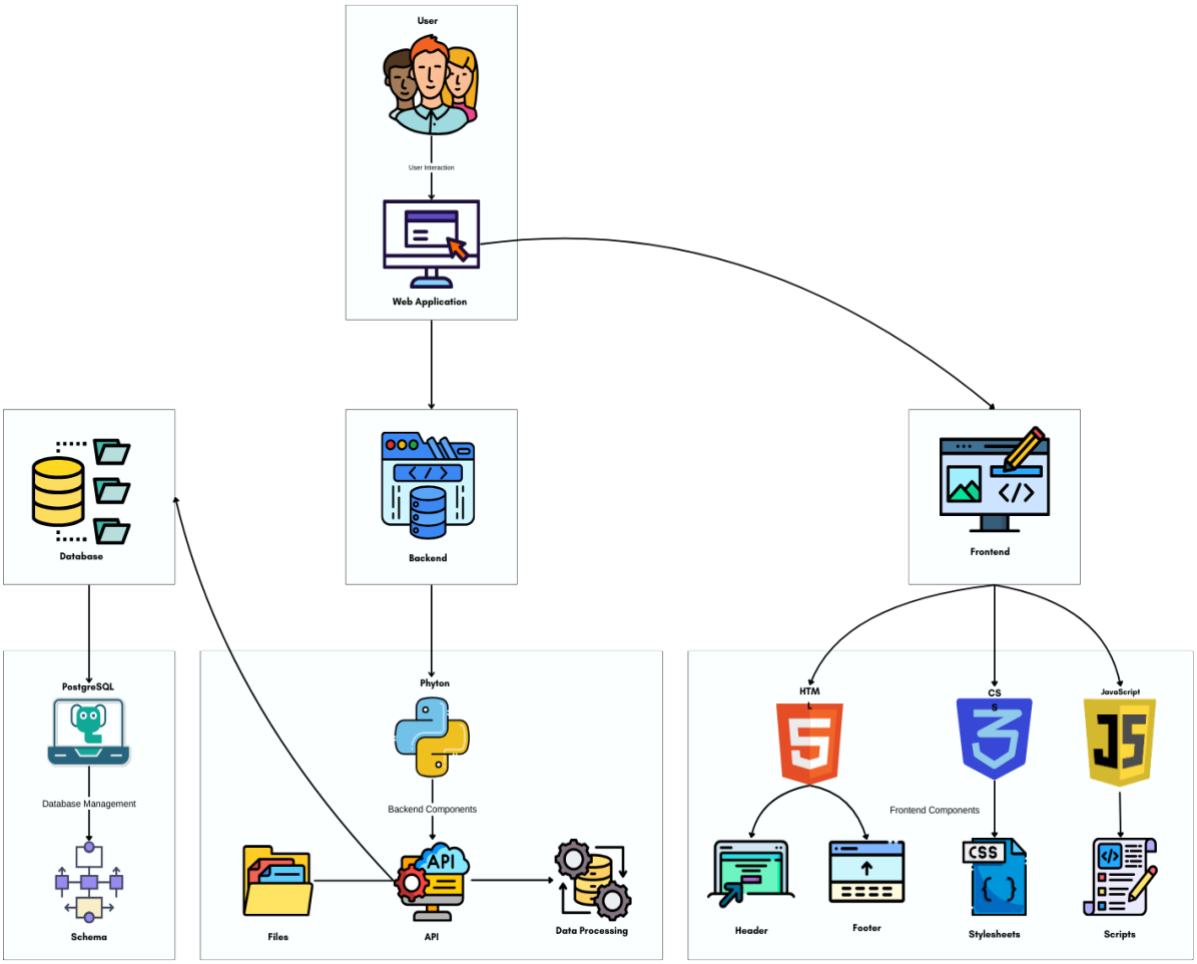


Figure 5 presents the technical architecture of the proposed system. It follows a modular client-server structure composed primarily of the user interface, frontend, backend, database, and analytics components. The user interacts with the system through the web application. The frontend uses HTML, CSS, and JavaScript, while the backend uses Python to handle application logic, API communication, files, and data processing. The database uses PostgreSQL for data storage and management. The processed information is then utilized by the analytics components to generate borrower risk, PPI, and scholarship-related insights.

Software Testing

Software testing ensures that the integrated predictive analytics system functions accurately, efficiently and in alignment with CARD, Inc.'s operational needs. Four essential testing methods were used:

Unit Testing focused on validating the functionality of individual system components, including preprocessing routines, risk classification modules, PPI prediction models and scholarship mechanisms. Each unit were tested using typical, boundary and error-prone inputs to ensure accuracy, stability and proper error handling. This step confirmed that the core components performed reliably before they were combined into the full system.

The unit testing process were conducted to verify the correctness and reliability of individual analytical components before system integration. A test-driven approach was applied using Pytest, where specific test cases were designed for key modules. Output was validated using assertion statements to ensure that results were not null, contained the required data fields, and followed the expected data structure.

Functions were implemented to assess each module, allowing issues to be identified at the component level. The tests were run in a controlled environment, with results automatically indicating pass or fail outcomes. This process ensured both functional accuracy and data integrity, confirming that outputs are reliable and suitable for further processing.

Integration Testing process were conducted to evaluate how the major components of the system function collectively within a batch upload–driven environment. The testing followed a scenario-based approach in which a complete dataset was uploaded through the centralized Batch Upload module, serving as the primary input for all integrated system functionalities.

During the testing period, predefined input data such as valid login credentials, uploaded CSV datasets, selected navigation actions, and report generation requests were used to verify the interaction between connected modules. The expected outputs described how the system should respond after processing the inputs. These outputs include successful user authentication, accessibility of all system modules, successful upload and processing of datasets, proper database storage of uploaded records, generation of PPI classifications and analytics, computation of risk scores and scholarship prioritization results, visualization of charts and dashboards, and generation of reports with accurate data.

The integration testing also verifies whether data flows correctly between modules without loss, delay, duplication, or processing errors. The uploaded dataset was traced throughout the system to ensure that each module correctly receives, processes, stores, and displays the required information. This includes validating whether PPI classifications, risk scores, scholarship priorities, and generated reports remain accurate and consistent across all connected modules. Furthermore, end-to-end workflow testing was performed to evaluate whether the

complete process from login, dataset upload, data processing, dashboard visualization, up to report generation, functions seamlessly within the integrated system environment.

Table 6
Integration Testing Test Cases

Test Case ID	Integrated Modules	Test Objective	Test Procedure	Expected Result	Actual Result	Status
IT-01	Login + Dashboard	Verify that the user can access the system after login	1. Enter valid username and password 2. Click login	User is redirected to the ppi dashboard		
IT-02	Login + Module Navigation	Verify that logged-in user can access all modules	1. Login 2. Navigate to PPI, Risk, Scholarship, Upload Dashboard	All modules are accessible without errors		
IT-03	Batch Upload + File Processing	Verify that CSV file can be uploaded and processed	1. Go to Batch Upload Dashboard 2. Select CSV file 3. Click Upload	File is uploaded and processed successfully with status "Success"		
IT-04	Batch Upload + Database Storage	Verify that uploaded data is stored in the database	1. Upload dataset 2. Check records in system modules	Uploaded data is stored and available across modules		
IT-05	Batch Upload + PPI Module	Verify that uploaded data generates PPI results	1. Upload dataset 2. Open PPI Dashboard	PPI score, classification (Pagtalubo/Regular), and analytics are displayed		
IT-06	Batch Upload + PPI Analytics Dashboard	Verify visualization of uploaded PPI data	1. Upload dataset 2. View PPI dashboard charts and map	Charts, total clients, classification distribution are correctly displayed		
IT-07	Batch Upload + PPI Likelihood Module	Verify prediction and likelihood results	1. Upload dataset 2. Open PPI Likelihood page	Predicted score, likelihood, and recommendations are shown		

IT-08	Batch Upload + PPI Classification Tables	Verify classification by Area, Unit, and Center	1. Upload dataset 2. Open classification tables	System groups clients correctly by branch, unit, and center		
IT-09	Batch Upload + At-Risk Processing	Verify risk scoring from uploaded data	1. Upload dataset 2. Open Risk Dashboard	Risk scores (Low, Moderate, High) are generated		
IT-10	At-Risk Module + Risk Dashboard	Verify visualization of risk analytics	1. Open Risk Dashboard 2. Check charts and trends	Risk distribution, trends, and top high-risk clients displayed		
IT-11	At-Risk Module + Client List	Verify detailed at-risk client records	1. Open Client At Risk module	Client list shows loan details, risk score, severity, and recommendation		
IT-12	Batch Upload + Risk Factors Overview	Verify computation of contributing risk factors	1. Upload dataset 2. Open Risk Factors Overview	Risk factors, severity levels, and percentages are displayed		
IT-13	Batch Upload + Scholarship Prioritization	Verify prioritization using uploaded data	1. Upload dataset 2. Open Scholarship Module	Eligible dependents, high priority, and total dependents are calculated		
IT-14	Scholarship Module + Visualization	Verify scholarship charts and distribution	1. Open Scholarship Dashboard	Education distribution and high-priority list are shown		
IT-15	Batch Upload + Cross-Module Consistency	Verify data consistency across modules	1. Upload dataset 2.Compare PPI, Risk, Scholarship outputs	Same client data is consistent across all modules		
IT-16	Batch Upload + Report Generation	Verify report generation from uploaded data	1. Upload dataset 2. Click Generate Report (PPI, Risk, Scholarship)	Reports are generated with correct data		
IT-17	End-to-End Workflow	Verify full system	1. Login 2. Upload dataset	Entire workflow		

		integration flow	3. View PPI 4. View Risk 5. View Scholarship	executes without errors		

Performance Testing process was conducted to evaluate the system's responsiveness, stability, and efficiency under normal operating conditions. The testing followed an endpoint-based approach, where key system functionalities such as data retrieval, analytics processing, and dashboard loading were executed and monitored. Each request was measured in terms of response time, processing speed, and system behavior during execution. The process involved repeatedly accessing different modules after batch data upload to observe how quickly the system responds and whether it maintains consistent performance across multiple operations. The results were then analyzed to determine whether response times fall within acceptable limits for real-time use and to identify areas needing optimization.

User Acceptance Testing (UAT) process was conducted to evaluate whether the developed system meets the functional requirements and is acceptable to its intended users under actual operational conditions. The testing followed a scenario-based approach in which selected end-users, particularly monitoring officers, will perform a series of predefined tasks using the system.

During the testing period, users executed specific actions such as entering valid login credentials, navigating through different system modules, selecting filter options, searching records using CID, uploading datasets, generating reports, and viewing dashboards and visualizations. The expected inputs include usernames and passwords for authentication, selected dropdown filters, search queries,

uploaded CSV files, navigation commands, and report generation requests. These inputs are expected to trigger corresponding system processes and outputs based on the defined functional requirements.

The expected outputs described how the system should respond after processing the given inputs and actions. These outputs include successful authentication and redirection to dashboards, display of summary information and analytics, retrieval of matching client records, updating of filtered data, generation of reports, visualization of maps and charts, and presentation of recommendations and classifications. The testing also verifies whether the system correctly displays PPI classifications, scholarship prioritization results, at-risk client records, uploaded file information, and user management details.

The UAT process aims to determine whether the system functions accurately, consistently, and responsively from the perspective of the intended users. Through the execution of the predefined test cases, the study will evaluate whether the system's actual behavior aligns with the expected outputs specified in the testing table and whether the developed system satisfies the operational needs and usability expectations of CARD, Inc. personnel.

Table 7
User Acceptance Testing Test Cases

Test Case ID	Test Scenario	Steps	Expected Result	Actual Result	Status
UAT-001	Verify user can log in using valid credentials	1. Open the system login page. 2. Enter valid username. 3. Enter valid password. 4. Click Log In.	The system should authenticate the user and redirect to the PPI dashboard.		
UAT-002	Verify PPI dashboard	1. Log in to the system. 2. Navigate	The dashboard should display the		

	displays summary information	to PPI > Progress Out of Poverty Index.	average PPI score, total clients, Pagtalubo count, Regular count, map, and trend chart.		
UAT-003	Verify PPI dashboard filter works properly	1. Open Progress Out of Poverty Index page. 2. Select a filter option from Filter by dropdown.	The displayed dashboard data should update according to the selected filter.		
UAT-004	Verify PPI Likelihood page displays client prediction records	1. Navigate to PPI > PPI Likelihood.	The system should display a table of client records with predicted score, predicted likelihood, poverty likelihood, and action recommendation.		
UAT-005	Verify search by CID in PPI Likelihood works	1. Open PPI Likelihood page. 2. Enter a valid CID in the search field.	The system should return the matching client record.		
UAT-006	Verify PPI Classification page displays area-based classification summary	1. Navigate to PPI > PPI Classification.	The system should show the list of branches/areas with active clients, Pagtalubo count, Regular count, and percentages.		
UAT-007	Verify Scholarship Prioritization dashboard displays summary cards	1. Navigate to Scholarship > Scholarship Prioritization.	The system should display total eligible dependents, high priority dependents, and total dependents.		
UAT-008	Verify Scholarship Prioritization map and chart are shown	1. Open Scholarship Prioritization page.	The system should display the geographic map and education level distribution chart.		
UAT-009	Verify high priority dependent list is displayed	1. Open Scholarship Prioritization page. 2. View the high priority dependents table.	The table should show the names of clients/dependents with area, unit, and center details.		
UAT-010	Verify Generate Report button for scholarship module works	1. Navigate to Scholarship Prioritization. 2. Click Generate Report.	The system should generate or prepare the scholarship report successfully.		

UAT-011	Verify At-Risk dashboard displays risk overview	1. Navigate to At-Risk > Risk Score Dashboard.	The system should display the map, top high-risk branches, top high-risk clients, percentage per risk level, and risk trend chart.		
UAT-012	Verify Client at Risk page displays client records by severity	1. Navigate to At-Risk > Client at Risk.	The system should display clients with loan details, arrears, risk score, risk severity, and recommended action.		
UAT-013	Verify filtering of Client at Risk records works	1. Open Client at Risk page. 2. Select a location from the filter dropdown.	The table should update and show records based on the selected location.		
UAT-014	Verify search by CID in Client at Risk works	1. Open Client at Risk page. 2. Enter a valid CID in the search box.	The system should return the matching at-risk client record.		
UAT-015	Verify Risk Factors Overview page displays trends and factors	1. Navigate to At-Risk > Risk Factor Overview.	The page should display monthly risk trend, at-risk loans chart, risk factor table, and recommended action section.		
UAT-016	Verify Batch Upload page displays uploaded files and report options	1. Navigate to Batch Upload.	The system should display uploaded file records, row counts, upload dates, status, and report generation options.		
UAT-017	Verify user management page displays user records	1. Navigate to User Management > Users.	The system should display the list of users with user ID, staff ID, username, name, position, and action buttons.		
UAT-018	Generate report	1. Navigate to Batch Upload 2. Select type of report > Generate	The system should open another page to generate report.		

Sampling Design. This study used a purposive sampling design, selecting 1–30 CARD, Inc. monitoring officers who represent the primary users of the

prescriptive analytics system. Because these personnel are directly involved in monitoring client risk and evaluating borrower information, they are the most suitable respondents for assessing the system's usability, acceptance, and operational relevance. This sampling approach ensures that feedback comes from individuals who are familiar with CARD, Inc.'s workflows, allowing the study to gather meaningful insights on both system usability and user acceptance. Since the goal of the evaluation is to assess the prototype's effectiveness rather than generalize findings to a broader population, the chosen sample size is appropriate and aligned with the study's purpose.

Data Collection Instrument. Data will be collected using a structured, researcher-made survey questionnaire composed of two parts: (1) a usability evaluation based on the Post-Study System Usability Questionnaire (PSSUQ), and (2) a Technology Acceptance Model (TAM)-based survey. The PSSUQ component is designed to assess users' experiences after interacting with the system, focusing on system usefulness, information quality, interface quality, ease of navigation, and overall satisfaction. The TAM component will evaluate user acceptance through key constructs such as Perceived Usefulness (PU) and Perceived Ease of Use (PEOU), which measure how the system enhances job performance and how easy it is to use. Both instruments will use a Likert-scale format to allow respondents to express their level of agreement in a standardized and measurable way. The combination of PSSUQ and TAM ensures a comprehensive evaluation of both usability and user acceptance.

Validation of the Questionnaire. The survey questionnaire will undergo content validation to ensure that each item accurately measures the intended constructs. The PSSUQ-based items were reviewed to confirm alignment with system usability dimensions, including system usefulness, information quality, and interface quality. Similarly, the TAM-based items were validated to ensure they effectively capture perceived usefulness and perceived ease of use.

Statistical Treatment. Descriptive statistics were used to analyze responses from both the PSSUQ-based usability questionnaire and the TAM-based survey. Frequency counts will summarize respondent profiles and categorical data, while mean scores and standard deviations were used to interpret user ratings across usability and acceptance dimensions. Specifically, mean scores were computed for PSSUQ subscales (System Usefulness, Information Quality, and Interface Quality) and TAM constructs (Perceived Usefulness and Perceived Ease of Use). Given the evaluative nature of the study and the relatively small sample size, descriptive methods are appropriate for identifying trends in usability and user acceptance without the need for inferential analysis. These results provide a clear understanding of how users perceive the system and identify areas for further improvement prior to wider implementation.

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